Modul 03 Basis Data

PRAKTIKUM BASIS DATA

Scripting Sql Basic Query

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SOAL TP 1

Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan

JAWAB

Worksheet | Query Builder

```
CREATE TABLE mahasiswa (
    id NUMBER PRIMARY KEY,
    nama VARCHAR2(100),
    tempat_lahir VARCHAR2(50),
    tanggal_lahir DATE,
    no_hp VARCHAR2(15),
    email VARCHAR2(100),
    tinggi_badan NUMBER,
    berat_badan NUMBER
);
```

Script Output | Task completed in 0.47 seconds

```
Table MAHASISWA created.
```

SOAL TP 2

Lakukan insert data pada table nomor 1 sebanyak 15 rows.

JAWAB

Worksheet | Query Builder

```
INSERT INTO mahasiswa VALUES (1,'Andi','Jakarta',DATE '2002-01-10','081234567890','andi@gmail.com',170,60);
INSERT INTO mahasiswa VALUES (2,'Budi','Bandung',DATE '2001-02-12','081234567891','budi@gmail.com',168,58);
INSERT INTO mahasiswa VALUES (3,'Citra','Surabaya',DATE '2002-03-15','081234567892','citra@gmail.com',160,50);
INSERT INTO mahasiswa VALUES (4,'Dina','Medan',DATE '2001-04-20','081234567893','dina@gmail.com',162,52);
INSERT INTO mahasiswa VALUES (5,'Eko','Semarang',DATE '2002-05-05','081234567894','eko@gmail.com',175,65);
INSERT INTO mahasiswa VALUES (6,'Fajar','Bogor',DATE '2001-06-18','081234567895','fajar@gmail.com',172,62);
INSERT INTO mahasiswa VALUES (7,'Gina','Depok',DATE '2002-07-22','081234567896','gina@gmail.com',158,48);
INSERT INTO mahasiswa VALUES (8,'Hadi','Bekasi',DATE '2001-08-30','081234567897','hadi@gmail.com',180,70);
INSERT INTO mahasiswa VALUES (9,'Intan','Padang',DATE '2002-09-10','081234567898','intan@gmail.com',165,55);
INSERT INTO mahasiswa VALUES (10,'Joko','Malang',DATE '2001-10-11','081234567899','joko@gmail.com',173,66);
INSERT INTO mahasiswa VALUES (11,'Kiki','Bali',DATE '2002-11-12','081234567800','kiki@gmail.com',160,50);
INSERT INTO mahasiswa VALUES (12,'Lina','Yogyakarta',DATE '2001-12-13','081234567801','lina@gmail.com',162,52);
INSERT INTO mahasiswa VALUES (13,'Mira','Solo',DATE '2002-01-14','081234567802','mira@gmail.com',159,49);
INSERT INTO mahasiswa VALUES (14,'Nanda','Lampung',DATE '2001-02-15','081234567803','nanda@gmail.com',174,67);
INSERT INTO mahasiswa VALUES (15,'Oki','Palembang',DATE '2002-03-16','081234567804','oki@gmail.com',171,64);
```

Script Output | Task completed in 0.421 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:

a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)

```
SELECT LOWER(nama) FROM mahasiswa;

SELECT UPPER(nama) FROM mahasiswa;

SELECT INITCAP(nama) FROM mahasiswa;

SELECT CONCAT(nama, ' - Mahasiswa') FROM mahasiswa;

SELECT nama, LENGTH(nama) FROM mahasiswa;

SELECT SUBSTR(nama,1,3) FROM mahasiswa;

SELECT LPAD(nama,10,'*') FROM mahasiswa;

SELECT TRIM('  ANDI  ') FROM dual;
```

```
SELECT INSTR(nama,'a') FROM mahasiswa;

SELECT REPLACE(nama,'a','@') FROM mahasiswa;
```

OUTPUT

```
LOWER(NAMA)
--------------------------------------------------------------------------------
andi
budi
citra
dina
eko
fajar
gina
hadi
intan
joko
kiki

LOWER(NAMA)
--------------------------------------------------------------------------------
lina
mira
nanda
oki

15 rows selected.

UPPER(NAMA)
--------------------------------------------------------------------------------
ANDI
BUDI
CITRA
DINA
EKO
FAJAR
GINA
HADI
INTAN
JOKO
KIKI

UPPER(NAMA)
--------------------------------------------------------------------------------
LINA
MIRA
NANDA
OKI

15 rows selected.
```

```
INITCAP(NAMA)
--------------------------------------------------------------------------------
Andi
Budi
Citra
Dina
Eko
Fajar
Gina
Hadi
Intan
Joko
Kiki

INITCAP(NAMA)
--------------------------------------------------------------------------------
Lina
Mira
Nanda
Oki

15 rows selected.
```

```
CONCAT(NAMA,'-MAHASISWA')
--------------------------------------------------------------------------------
Andi - Mahasiswa
Budi - Mahasiswa
Citra - Mahasiswa
Dina - Mahasiswa
Eko - Mahasiswa
Fajar - Mahasiswa
Gina - Mahasiswa
Hadi - Mahasiswa
Intan - Mahasiswa
Joko - Mahasiswa
Kiki - Mahasiswa

CONCAT(NAMA,'-MAHASISWA')
--------------------------------------------------------------------------------
Lina - Mahasiswa
Mira - Mahasiswa
Nanda - Mahasiswa
Oki - Mahasiswa

15 rows selected.
```

```
NAMA
--------------------------------------------------------------------------------
Andi
Budi
Citra
Dina
Eko
Fajar
Gina
Hadi
Intan
Joko
Kiki

NAMA
--------------------------------------------------------------------------------
Lina
Mira
Nanda
Oki

15 rows selected.

SUB
---
And
Bud
Cit
Din
Eko
Faj
Gin
Had
Int
Jok
Kik

SUB
---
Lin
Mir
Nan
Oki

15 rows selected.
```

```
LPAD(NAMA,
----------
******Andi
******Budi
*****Citra
******Dina
*******Eko
*****Fajar
******Gina
******Hadi
*****Intan
******Joko
******Kiki

LPAD(NAMA,
----------
******Lina
******Mira
*****Nanda
*******Oki

15 rows selected.

TRIM
----
ANDI


INSTR(NAMA,'A')
---------------
              0
              0
              5
              4
              0
              2
              4
              2
              4
              0
              0

INSTR(NAMA,'A')
---------------
              4
              4
              2
              0

15 rows selected.
```

```
REPLACE(NAMA,'A','@')
--------------------------------------------------------------------------------
Andi
Budi
Citr@
Din@
Eko
F@j@r
Gin@
H@di
Int@n
Joko
Kiki

REPLACE(NAMA,'A','@')
--------------------------------------------------------------------------------
Lin@
Mir@
N@nd@
Oki

15 rows selected.
```

b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))

```
SELECT SYSDATE FROM dual;

SELECT ADD_MONTHS(SYSDATE,3) FROM dual;

SELECT LAST_DAY(SYSDATE) FROM dual;

SELECT NEXT_DAY(SYSDATE,'MONDAY') FROM dual;

SELECT MONTHS_BETWEEN(SYSDATE, tanggal_lahir) FROM mahasiswa;

SELECT ROUND(SYSDATE,'MONTH') FROM dual;

SELECT ROUND(SYSDATE,'YEAR') FROM dual;
```

OUTPUT

```
SYSDATE
---------
05-MAR-26


ADD_MONTH
---------
05-JUN-26


LAST_DAY(
---------
31-MAR-26


NEXT_DAY(
---------
09-MAR-26


MONTHS_BETWEEN(SYSDATE,TANGGAL_LAHIR)
-------------------------------------
                           289.870117
                           300.805601
                           287.708827
                           298.547536
                                  286
                           296.612052
                            283.48302
                           294.224956
                           281.870117
                           292.837859
                           279.805601

MONTHS_BETWEEN(SYSDATE,TANGGAL_LAHIR)
-------------------------------------
                           290.773343
                           289.741085
                           300.708827
                           287.676568

15 rows selected.


ROUND(SYS
---------
01-MAR-26


ROUND(SYS
---------
01-JAN-26
```

C. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

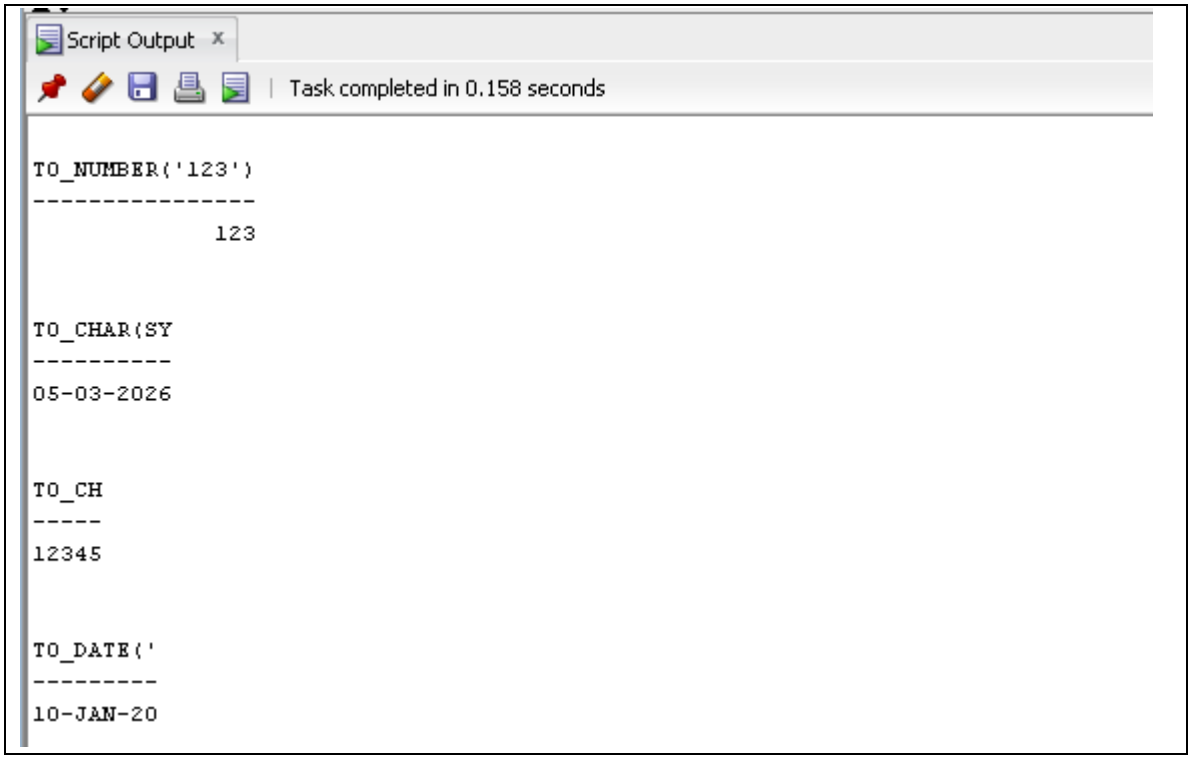
```
SELECT TO_NUMBER('123') FROM dual;

SELECT TO_CHAR(SYSDATE,'DD-MM-YYYY') FROM dual;

SELECT TO_CHAR(12345) FROM dual;

SELECT TO_DATE('10-01-2020','DD-MM-YYYY') FROM dual;
```

OUTPUT



D. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

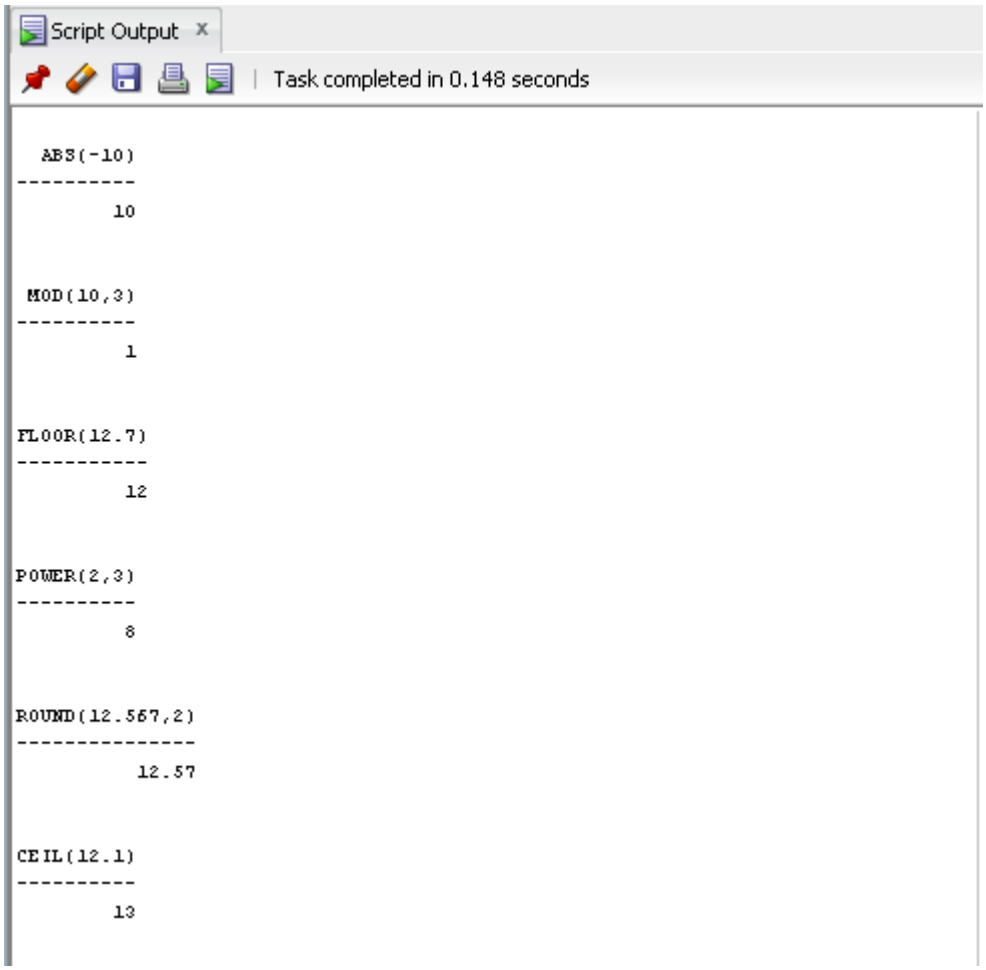
```
SELECT ABS(-10) FROM dual;

SELECT MOD(10,3) FROM dual;

SELECT FLOOR(12.7) FROM dual;

SELECT POWER(2,3) FROM dual;
```

SELECT ROUND(12.567,2) FROM dual;

SELECT CEIL(12.1) FROM dual;

OUTPUT